Ryan Qian 1010857663

Kaden Seto 1010859493

Jeffery Tian 1010869945

Jadon Tsai 1010870283

September 10 3:00-6:00 PM

Path Independence and Tank Volume from Ideal Gas Measurements

1 Abstract

In the first part of this experiment, we connected two air tanks and observed how pressure and temperature changed as the tanks reached equilibrium in two distinct ways. By showing that the steady-state pressure and temperature of air, which is modeled as an ideal gas, are the same for both processes, we demonstrate the path independence of pressure and temperature. In the second part of the experiment, we apply the Ideal Gas Law to determine the volume of the gas tank after a known quantity of gas is added. The experiments conducted in the lab confirmed the path independence of pressure within 4% and temperature within 0.5% and determined tank volume to be 7.17 litres, with a volume ratio between the two tanks of 1.74. These results are consistent with the concept of path independence and ideal gas behaviour for atmospheric air at moderate temperatures and pressures.

2 Introduction

Path independence is the concept that the equilibrium values of certain properties in a changing thermodynamic system do not depend on the mechanism of change. We call these properties state properties. We can also model the atmosphere at moderate pressure and temperature using the ideal gas model, which models gas molecules as point particles that only interact with other molecules in collisions, ignoring long range interactions between molecules observed in reality. We can thus use the Ideal Gas Law to express a relationship between the values of an ideal gas with the following equation:

$$PV = mRT \tag{1}$$

where P is pressure in Pascals (Pa), V is volume (m^3), m is the mass of the gas (kg), T is the absolute temperature in Kelvin (K), and R is the specific gas constant ($\frac{J}{kg*K}$).

The first objective of this lab is to show that the steady-state thermodynamic properties of a system are path-independent by showing that Ideal Gas Law remains valid through time by conducting two equilibrium processes on the pressure of the system, each with demonstrably different paths. The second objective aims to apply the Ideal Gas Law to compute and determine the volume of the larger gas tank, which will use the ratio determined from the first experiment to calculate the volume of the smaller tank. The analyzed results that will be discussed in this lab will provide clear evidence to demonstrate that the Ideal Gas model can be applied to atmospheric air at moderate pressure and temperature.

3 Method

3.1 Apparatus

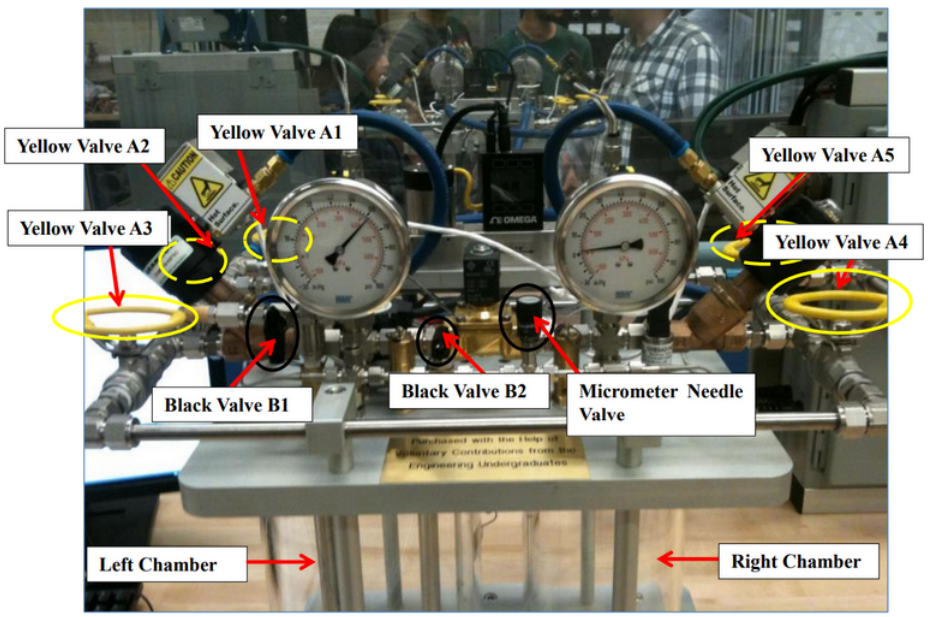

Figure 1: A picture of the experimental setup, with important valves labeled. [2]

The lab setup (Figure 1) consisted of two connected air tanks, connected via a system of valves, importantly a solenoid and micrometer needle valve, and a mass flow controller operated with LabVIEW, a software system used to facilitate data collection and processing. Pressure and temperature were measured with an integrated manometer and thermocouples respectively. A mass flow controller also logged the flow rate of air entering the system.

3.2 Procedure

3.2.1 General Procedure for Parts 1 and 2

Figure 2: A picture of the manometer and ambient thermometer present in the lab

These steps must be performed first for both parts of the lab. Procedures specific to each part will be described in 3.2.2 and 3.2.3. Note that in Figure 1, the left tank and right tank are denoted by left chamber and right chamber, respectively.

1. Record the ambient pressure and temperature using the manometer and thermometer in the lab (Figure 2).
2. Begin data collection by pressing "Start Collecting Data".
3. Set the pressure in the left tank to 40 psi, ensuring that the airflow is increased incrementally to avoid damage to the system.
4. Proceed with the specific procedures associated with each part.

3.2.2 Specific Procedure for Part 1

In Part 1, we allow both tanks to come to equilibrium, first by immediately opening the center solenoid valve (steps 2-3), then by gradually opening the micrometer needle valve (steps 5-6).

1. After following the procedure outlined in 3.2.1, vacuum the right tank to -6 psi.
2. Open the center solenoid, then leave it open until there is a negligible temperature difference between the tanks.
3. Stop recording and save the data to a file before resetting the system.
4. Repeat the process outlined in 3.2.0 and vacuum the right tank to -6 psi.

5. Open valve B2 and the micrometer valve, then leave both open until there is a negligible pressure difference between the tanks.

6. Stop recording and save the data to a file before resetting the system and closing both valves, then empty both tanks.

3.2.3 Specific Procedure for Part 2

In Part 2, we wait for the left tank to reach equilibrium, while recording the mass flow rate and the final pressure and temperature of the gas.

1. After following the procedure outlined in 3.2.0, wait for the pressure and temperature to stabilize.

2. Stop recording and save the data to a file before emptying the left tank and resetting the system.

4 Results and Discussion

4.1 Formulae for Figures

For Figures 3 and 4, the absolute pressure was plotted, and determined from the gauge pressure using the following formula, where $P_{atmospheric}$ was measured to be 0.1013 MPa and all pressures are in MPa:

$$P_{abs} = P_{Gauge} \times 0.006894 + P_{atmospheric} \tag{2}$$

For Figures Z and A, the temperature was expressed in Kelvin, and determined from the gauge reading using the formula:

$$T_{Kelvin} = T_{Celsius} + 273.15 \tag{3}$$

Vertical axis cutoffs for all figures were determined by visual inspection for presentation clarity.

4.2 Demonstrating Path Independence of Pressure and Temperature

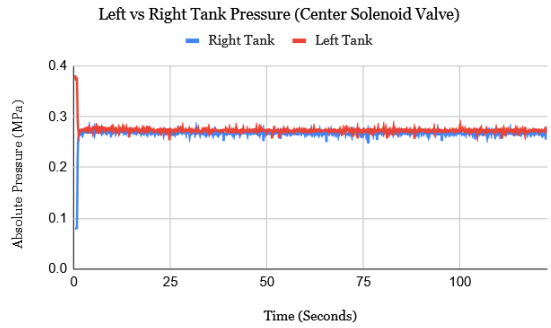

Figure 3: Pressure vs. Time for the Center Solenoid. Pressure was converted to absolute pressure MPa.

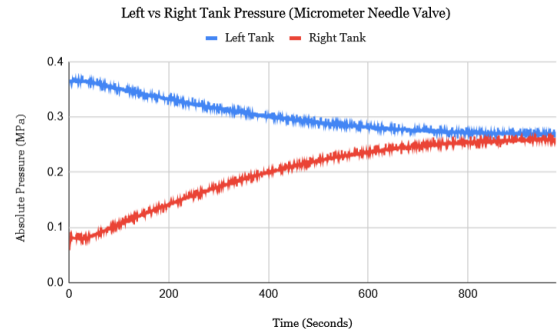

Figure 4: Pressure vs. Time for the Micrometer Valve. Pressure was converted to absolute pressure MPa.

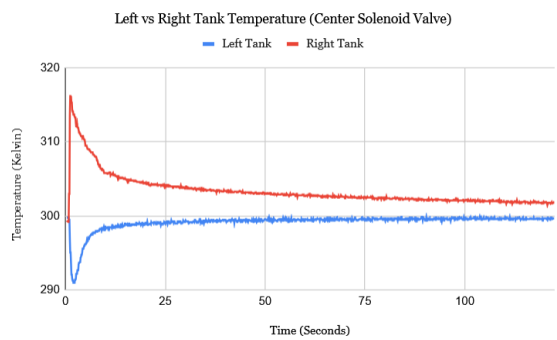

Figure 5: Temperature vs. Time for the Center Solenoid. Temperature was converted to absolute degrees Kelvin.

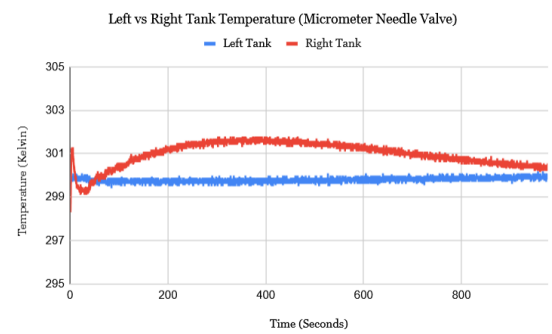

Figure 6: Temperature vs. Time for the Micrometer Valve. Temperature was converted to absolute degrees Kelvin.

Our first experiment demonstrates the path independence of thermodynamic properties as the equilibrium pressures and temperatures of both tanks remain constant despite the difference in process. When the center solenoid was opened, the tanks equalized in pressure within 3 seconds (Figure 3), while the temperature of the right tank increased dramatically (Figure 5). In comparison, when the micrometer valve was opened, it took around 8 minutes for the pressures to equalize (Figure 4), while the temperature of the right tank decreased before increasing again (Figure 6). It is clear that the two thermodynamic processes are distinct; however, the final equilibrium pressures and temperatures are within 4% and 0.5% of one another, respectively. Thus, we have demonstrated that the final thermodynamic properties of this system remain constant regardless of the process.

Process Variable	$P_{initial}$ of Part A (MPa± 0.0000005 MPa)	P_{final} of Part A (MPa± 0.0000005 MPa)	P_{final} of Part B (MPa ± 0.0000005 MPa)	P_{final} % Difference	$T_{initial}$ of Part A (K± 0.05 K)	T_{final} of Part A (K± 0.05 K)	T_{final} % Difference
Tank 1 Value	0.3811964	0.2729606	0.2695136	1	299.75	300.05	0.1%
Tank 2 Value	0.0799286	0.2688242	0.2591726	4	301.75	300.35	0.5%

Table 1: Final values measured from graphs

4.3 Determining the Volume Ratio Between the Two Tanks

In the first experiment, the system consisting only of both tanks is a closed system when all valves are closed so that no mass of air can enter or exit. As a result, due to conservation of mass, the total amount of gas present within both tanks remains constant throughout any thermodynamic process. Using this fact with the Ideal Gas model yields a final equation for the volume ratio between the two tanks:

$$\frac{V_1}{V_2} = \frac{T_{2f}T_{2i}(P_{2f}T_{2i} - P_{2i}T_{2f})}{T_{1f}T_{1i}(P_{1i}T_{1f} - P_{1f}T_{1i})} \tag{4}$$

We used this equation to find that the final volume ratio was 1.74 ± 0.01. Note that we used Part A pressure and temperature data from Table 1 for this calculation, as the system remained at equilibrium for a longer duration in Part A.

4.4 Finding the Volume of the Larger Tank

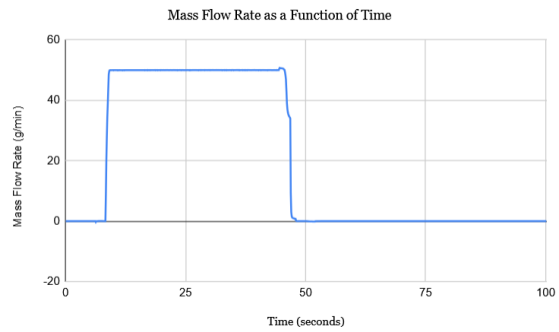

Figure 7: Mass Flow Rate vs. Time for the Left Tank. We approximated the integral as the area of a rectangle but it can be clearly seen that the area is greater than a rectangle.

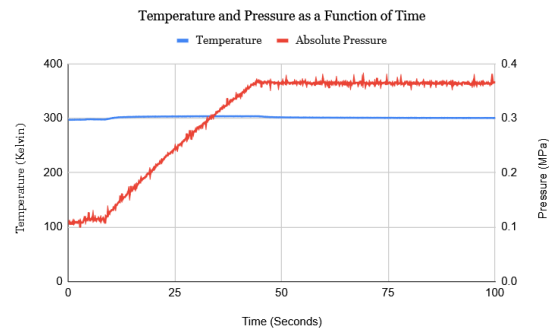

Figure 8: Temperature and Pressure vs. Time for the Left Tank. Temperature was converted to absolute degrees Kelvin. Pressure was converted to absolute pressure MPa.

Process Variable	T_{final} (K)	P_{final} (MPa)	Mass of Gas (g)	Volume of Gas (L)
Tank 1 Value	300.65 ± 0.05	0.2647296 ± 0.0000005	30.4 ± 0.4 g	7.17 ± 0.1 L

Table 2: A table showing the final thermodynamic properties and the mass of the gas in the Left Tank, as well as our calculated volume.

The total mass in the chamber was estimated by multiplying the mass flow rate of 50 $\frac{g}{min}$ by the duration the valves were kept open, around 36.5 seconds, as we estimated the region of the mass flow rate graph when the mass flow rate was 50 $\frac{g}{min}$ as rectangular (Figure 7). The mass was estimated to be 30.4 grams. We can see from Figure 8 that the pressure and temperature reach constant equilibrium values, so that we can use the ideal gas equation the equation, $V = \frac{MRT}{P}$, (where the R for air was used), and the tank volume was calculated to be 7.17 litres.

4.5 Physical Sources of Uncertainty

As all values recorded in this experiment were done by digital measurements, there was no influence in human measurement uncertainties. To allow for error within machine calibration, we opted to include a ±5 uncertainty on the last significant digit of each recorded value.

Our first experiment demonstrates the path independence of thermodynamic properties as the equilibrium pressures and temperatures of both tanks remain constant despite the difference in process. When the center solenoid was opened in the fast equalization method, both tanks equalized in pressure very quickly, while the temperature of the right tank increased dramatically. This was a quasi-adiabatic process, since there was very little time for heat transfer between the gas and surroundings, thus increasing the temperature of

the gas. The energy transferred was mostly work rather than heat. In comparison, when the micrometer valve was opened in the slow equalization process, it took a significant amount of time for the pressures to equalize, while the temperature of the right tank decreased before increasing again, but by a relatively small amount compared to the fast process. This was indicative of a quasi-isothermal process, where heat was exchanged with the surroundings that acted as a heat reservoir. The fact that the isothermal process was modeled as a closed system compared to the adiabatic process that was closer to an isolated system meant that we could expect a lower total energy and therefore temperature of the closed system. There was also some heat transfer to surroundings with lack of insulation in the rest of the system, especially since the metal pipes heated up. Most of the energy was thus transferred as heat instead of work. Despite these drastic differences in pressure and temperature behavior, the equilibrium pressures (0.268 MPa average for the center solenoid, 0.268 MPa average for the micrometer valve) and temperatures (300.5 K average for the center solenoid and 300.0 K average for the micrometer valve) are within the uncertainties for both values, demonstrating path independence of these state properties.

4.6 Approximations Made in Calculations

The numerical integration we performed underestimates flow because opening and closing of valves was not instantaneous and there is also friction in the system meaning that some air got in before and after the peak value, meaning that the flow rate curve is closer to a trapezoid.

Although air is not an ideal gas, in this experiment we made the assumption that the compressibility of gas is negligible. The Ideal Gas model has two assumptions where it assumes that molecules have no volume and negligible intermolecular interactions, which are factors that have a negligibly small effect at moderate/room temperature and pressures, which the experiments were performed at. Therefore, air compressibility can be considered negligible and other sources of error were much more significant.

5 Conclusion

The results from this set of experiments supported the concept of path independence of pressure and temperature, as well as the ideal gas law under relatively moderate temperature and pressures. We showed that two different paths of reaching equilibrium between two tanks, quickly via the solenoid valve and slowly via the micrometer, reached the same final pressure within 4% error and same final temperature within 0.5% error, agreeing with the theory of path independence of thermodynamic state properties. We also found that the volume of a gas tank was 7.17 ± 0.1 L, which was found by measuring the mass of air entering the tank by integrating under the flow rate curve. The volume ratio between the two tanks was 1.7434.

6 References

1. CHE260 "CHE 260 Lab 1 - Ideal Gas Law" 2025a
2. CHE260 "CHE 260 - Introductory Manual" 2025b
3. CHM 363: Physical Chemistry 1 (Grinnell College) 2025